

20. This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13-17 students, regular-sized classes with 22-25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools.

a. Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?

Effect of small classes: The coefficient on small is positive and statistically significant ($p < 0.01$). Students in small classes score about 5.82 points higher on reading tests compared with students in regular classes, holding other factors constant.

Therefore, the results suggest that smaller classes improve reading performance.

Effect of teacher aides: The coefficient on aide is positive but statistically insignificant ($p = 0.391$). Having a teacher's aide does not significantly improve students' reading scores.

Thus, there is no strong evidence that teacher aides increase reading achievement.

Effect of teacher experience: The coefficient on tchexper is positive and highly significant. Students taught by more experienced teachers tend to achieve higher reading scores.

Specifically, one additional year of teaching experience is associated with about a 0.49-point increase in reading score.

Gender effect: The coefficient on boy is negative and statistically significant. Boys score about 6.16 points lower than girls on reading tests, on average.

This indicates a significant gender difference in reading performance.

Race effect: The coefficient on white_asian is positive and significant. White and Asian students score about 3.91 points higher on reading tests than other students, holding other variables constant.

Socioeconomic background: The coefficient on freelunch is negative and highly significant. Students receiving free lunch score about 14.77 points lower on reading tests.

This suggests that students from lower-income families tend to perform worse academically.

Overall model fit: The overall regression is statistically significant because the F-statistic has a p-value below 0.01. However, the R^2 is about 0.097, meaning that the model explains roughly 9.7% of the variation in reading scores.

Overall, the results suggest that class size, teacher experience, gender, race, and socioeconomic background all have important effects on students' reading performance, while the presence of a teacher's aide does not significantly improve scores.

b. Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [Hint: specify *SCHID* as the cross-section identifier and ID as the “time” identifier.]

	Estimate(OLS)	Estimate(FE)
<i>SMALL</i>	5.80***	6.49***
<i>AIDE</i>	0.82	0.996
<i>TCHEXPER</i>	0.49***	0.29***
<i>BOY</i>	-6.16***	-5.46***
<i>WHITE_ASIAN</i>	3.91***	8.03***
<i>FREELUNCH</i>	-14.77***	-14.59***

After including school fixed effects, the main conclusions remain largely unchanged. The coefficient on *SMALL* remains positive and statistically significant, suggesting that students in small classes continue to perform better on reading tests. The coefficient on *AIDE* is still statistically insignificant, indicating that having a teacher’s aide does not significantly improve reading performance. The coefficients on *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH* remain statistically significant and keep the same signs as in the OLS model, although some magnitudes change slightly after controlling for school-specific effects.

Overall, including school fixed effects controls for unobserved differences across schools and improves the model fit, but it does not substantially change the main conclusions from part (a).

c. Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?

The F-statistic is 16.698 with a p-value $< 2.2e-16$. Since the p-value is extremely small, we strongly reject the null hypothesis. This indicates that school fixed effects are jointly statistically significant. Therefore, unobserved school-specific characteristics play an important role in explaining variation in student reading scores.

Fixed effects have little impact when the explanatory variables are uncorrelated (or only weakly correlated) with unobserved school characteristics. In particular, if class assignment is random within schools and most variation in regressors occurs within schools rather than between schools, then adding fixed effects mainly absorbs school-level differences without significantly changing the estimated coefficients of other variables.

d. Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.

The random-effects estimates are very similar to those from the OLS and fixed-effects models. The coefficient on *small* remains positive and significant, while *aide* remains insignificant. Other variables keep the same signs and significance, with only minor changes in magnitude.

Some variables such as *white_asian*, *freelunch*, and possibly *tchexper* may be correlated with school-specific effects, since schools differ in socioeconomic composition and teacher quality. In contrast, *small* is less likely to be correlated with school effects due to random assignment within schools.

The Breusch–Pagan LM test strongly rejects the null of no random effects ($p < 0.01$), indicating that random effects are present and the RE model is preferred over pooled OLS.

e. Using the t-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?

Using the t-test in equation (15.36), we compare the fixed effects and random effects estimates. For *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*, the differences between FE and RE estimates are very small and not statistically significant at the 5% level. This suggests that the random effects estimator is consistent and that school-specific effects are not strongly correlated with these regressors.

Applying the same test to *BOY* also yields no significant difference between FE and RE estimates, indicating that *BOY* is also uncorrelated with school-specific effects.

f. Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

The Mundlak approach includes school averages of the explanatory variables to test whether the regressors are correlated with unobserved school effects.

The results show that several school-average variables, such as *small_bar*, *tchexper_bar*, *boy_bar*, and *white_asian_bar*, are statistically significant. This indicates that the explanatory variables are correlated with school-level unobserved heterogeneity.

Therefore, the random effects assumption is violated, and the fixed effects model is more appropriate.